Robin Lange

Research in robotics and intelligent systems

Autonomy, mission planning, and ground robotics

Collaborative autonomy / IHMC

Built a Rust/Python autonomy platform with Joint Activity Graphs and Zenoh. Integrated Spot robots, motion capture, hybrid simulation, and runtime safeguards.

Demonstrated collection and delivery with two Spot robots across 10+ hardware trials.



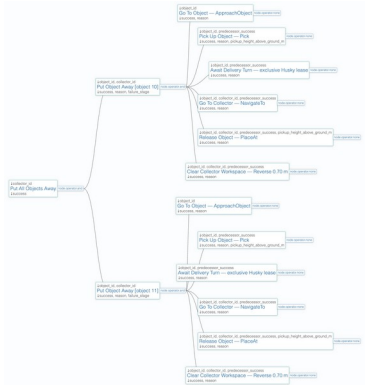
IHMC hardware demonstration



Green robot testing

Mission planning / Fraunhofer IVI

Implemented ROS 2 behavior trees and guarded state machines. Tested emergency interruption in simulation.



Joint Activity Graph

Ground robotics

Collaborative ROS 2 development and testing on Eagle Bot. TurtleBot4/Ouster integration for LiDAR SLAM and Nav2.

IHMC project feature

Research code

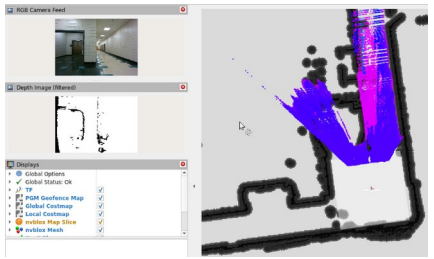
Perception and experimental systems

Visual SLAM, LiDAR hand control, and YOLO perception

UAV navigation

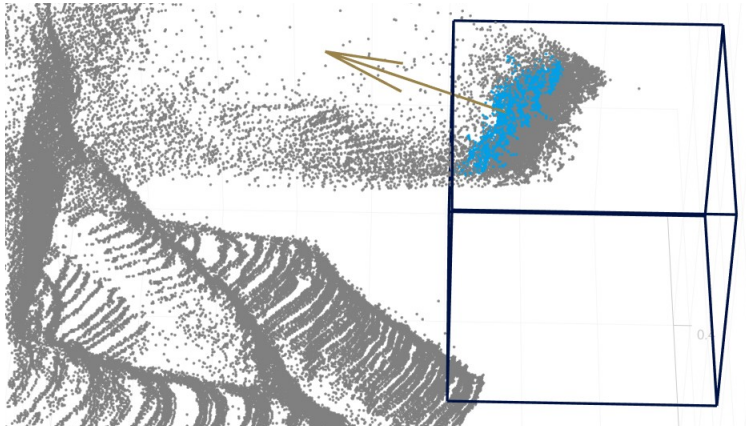
Integrated Jetson/RealSense visual SLAM and Nav2. A 110 s cart trial reached 14.5 cm RMSE on valid reference intervals.

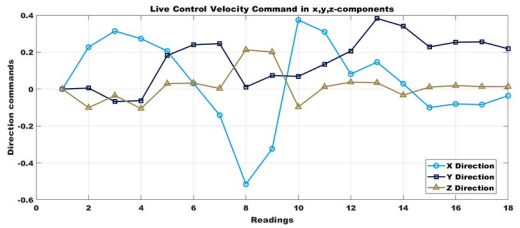




LiDAR hand control

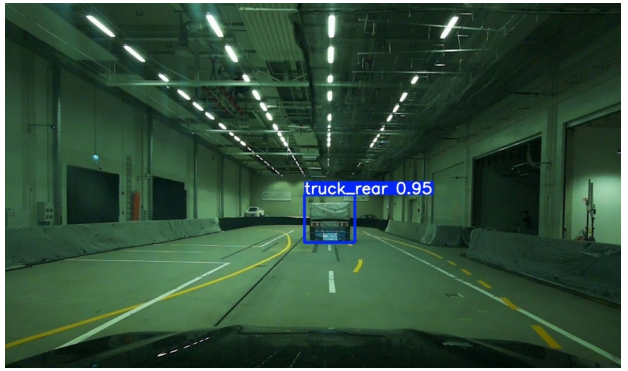
Tracked hand motion in MATLAB using LiDAR point clouds and sent velocity commands to a Tello UAV. Demonstrated hand-controlled flight.

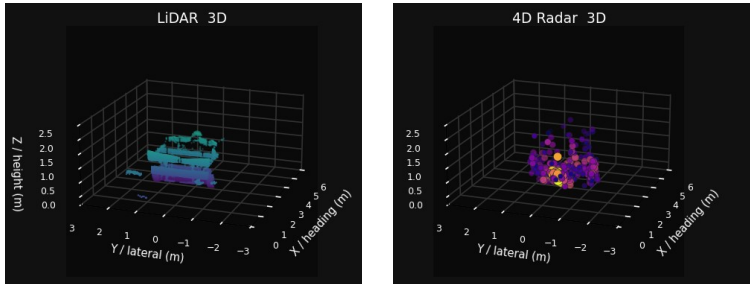




Sensor testing / CARISSMA

Summer internship: built soft crash targets and compared LiDAR/radar. Related work: fine-tuned YOLO for truck-rear detection.





Research outputs and current direction

Publications, collaborative studies, and an ongoing honors thesis

Published research / Co-author

LiDAR lane-keeping assistance

Rule-based oncoming-vehicle tracking, evaluated on nine real driving scenarios. SAE technical paper.

Data-driven simulation calibration

LSTM actuator modeling from wheel-encoder data. The study reports 36.3% lower wheel-velocity RMSE.

Earlier collaborative studies

LiDAR/camera detection with JPDA tracking.
LSTM fusion of LiDAR and ultrasonic range data.

Honors thesis / In progress

Learning-based predictive control for vehicle following

Combines GRU lead-vehicle prediction with constrained MPC and a shared braking supervisor in MATLAB/AVL. Investigates reliability-gated preview through matched simulation runs.

Additional research outputs

Warehouse platooning

SSRN preprint, co-author

Soft truck-underride target testing

Manuscript in preparation, lead author